

Worksheet # 9

Week 5, Day 1: Elements of visualization, choosing a plot, correlation, perception

Name: _____

NetID: _____

Date: _____

Guided checkpoints (optional)

1. Bertin's Seven Visual Variables

Name all seven.

2. Munzner's Magnitude Channels

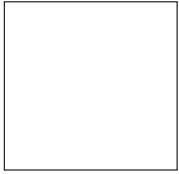
Rank the first five magnitude channels

3. Choosing a Plot: One Quantitative Feature

Name three plots for a single quantitative feature.

4. Pearson's Correlation Coefficient

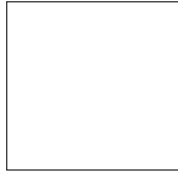
Sketch a scatter plot for each panel.



$r \approx 0.9$



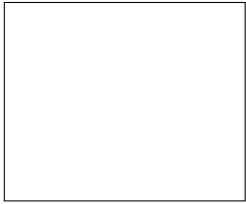
$r \approx 0$



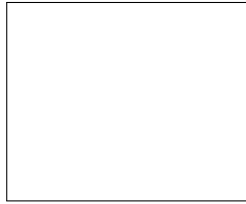
$r \approx -0.5$

5. Simpson's Paradox

Sketch the same data twice, with trends.



pooled



split by group

6. Choosing a Plot: Two Variables

Name and sketch a plot for visualizing data with **two categorical** variables. And for **one quantitative and one categorical** variable.

7. The Principle of Proportional Ink

State the principle.